

A IRLS

Algorithm 1 Iteratively Reweighted Least Squares (IRLS) for GLM Fitting

Require: $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$, link function $g(\cdot)$, variance function $V(\mu)$, tolerance $\epsilon > 0$, maximum iterations $T_{\max}$

Ensure: Estimated coefficients $\hat{\beta} \in \mathbb{R}^p$

```

1: Initialize: Set  $\beta^{(0)}$ 
2: Compute initial linear predictors:  $\eta^{(0)} \leftarrow X\beta^{(0)}$ 
3: Compute initial fitted means:  $\mu^{(0)} \leftarrow g^{-1}(\eta^{(0)})$ 
4: Set iteration counter:  $t \leftarrow 0$ 
5: repeat
6:   // Step 1: Compute weights
7:   for  $i = 1$  to  $n$  do
8:      $w_i^{(t)} \leftarrow \frac{1}{V(\mu_i^{(t)})} \left( \frac{d\mu_i}{d\eta_i} \Big|_{\mu_i^{(t)}} \right)^2 = \frac{1}{V(\mu_i^{(t)}) \cdot [g'(\mu_i^{(t)})]^2}$ 
9:   end for
10:   $W^{(t)} \leftarrow \text{diag}(w_1^{(t)}, \dots, w_n^{(t)})$ 
11:  // Step 2: Compute working response
12:  for  $i = 1$  to  $n$  do
13:     $z_i^{(t)} \leftarrow \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \cdot g'(\mu_i^{(t)})$ 
14:  end for
15:  // Step 3: Solve weighted least squares
16:   $\beta^{(t+1)} \leftarrow (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}$ 
17:  // Step 4: Update fitted values
18:   $\eta^{(t+1)} \leftarrow X\beta^{(t+1)}$ 
19:   $\mu^{(t+1)} \leftarrow g^{-1}(\eta^{(t+1)})$ 
20:  // Step 5: Check convergence
21:   $\Delta \leftarrow \|\beta^{(t+1)} - \beta^{(t)}\|_2$ 
22:   $t \leftarrow t + 1$ 
23: until  $\Delta < \epsilon$  or  $t \geq T_{\max}$ 
24: if  $t \geq T_{\max}$  then
25:   Warning: Algorithm did not converge
26: end if
27: return  $\hat{\beta} \leftarrow \beta^{(t)}$ 

```

Convergence criteria. Common stopping rules include:

- Parameter change: $\|\beta^{(t+1)} - \beta^{(t)}\|_2 < \epsilon$ (typically $\epsilon = 10^{-6}$).
- Deviance change: $|\mathcal{D}(\beta^{(t+1)}) - \mathcal{D}(\beta^{(t)})| < \epsilon$, where $\mathcal{D}(\beta) = -2\ell(\beta)$.
- Gradient norm: $\|s(\beta^{(t+1)})\|_2 < \epsilon$.

When using the IRLS algorithm, there are some common practical considerations:

- Starting values: Good initialization accelerates convergence. Common choices: $\beta^{(0)} = 0$ or fit from linear regression.
- Numerical stability: Add small ridge λI to $X^\top W X + \lambda I$ if matrix is near-singular.

B Custom Logistic Regression Validation

Following figures show the key points of the Custom Logistic Regression Validation:

```
> summary(full_model)

Call:
glm(formula = target ~ age + sex + cp + trestbps + chol + fbs +
     restecg + thalach + exang + oldpeak + slope + ca + thal,
     family = binomial, data = data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.739e+01  1.017e+01  -2.693  0.00707 **
age          -1.180e-02  2.517e-02  -0.469  0.63923
sex1         1.482e+00  5.285e-01  2.803  0.00506 **
cp2          8.306e-01  7.655e-01  1.085  0.27790
cp3          7.025e-02  6.523e-01  0.108  0.91423
cp4          1.916e+00  6.583e-01  2.910  0.00361 **
trestbps     3.183e+00  1.669e+00  1.906  0.05659 .
chol         1.655e+00  1.197e+00  1.383  0.16680
fbs1        -4.667e-01  6.013e-01  -0.776  0.43764
restecg1     3.812e-01  2.932e+00  0.130  0.89654
restecg2     4.746e-01  3.857e-01  1.230  0.21852
thalach     -4.024e-05  2.394e-05  -1.680  0.09288 .
exang1       7.283e-01  4.528e-01  1.609  0.10769
oldpeak      4.385e-01  2.392e-01  1.833  0.06684 .
slope2       9.589e-01  4.689e-01  2.045  0.04084 *
slope3       8.509e-01  9.115e-01  0.933  0.35058
ca           1.228e+00  2.783e-01  4.413  1.02e-05 ***
thal6       -1.099e-01  7.947e-01  -0.138  0.88998
thal7        1.408e+00  4.339e-01  3.244  0.00118 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 395.69  on 286  degrees of freedom
Residual deviance: 186.82  on 268  degrees of freedom
AIC: 224.82

Number of Fisher Scoring iterations: 6
```

(a) Full model summary - R

Variable	Coef	Std.Err	z	P> z	[0.025	0.975]	
Intercept	-26.7479	9.7965	-2.730	0.0063	-45.9486	-7.5472	**
trestbps	2.8936	1.5842	1.827	0.0678	-0.2113	5.9985	.
chol	1.7062	1.1540	1.479	0.1393	-0.5555	3.9680	.
thalach	-0.0000	0.0000	-1.740	0.0818	-0.0001	0.0000	.
oldpeak	0.4397	0.2318	1.897	0.0578	-0.0145	0.8940	.
ca	1.1767	0.2606	4.516	0.0000	0.6660	1.6874	***
cp_2.0	0.7906	0.7589	1.042	0.2975	-0.6967	2.2780	
cp_3.0	0.0465	0.6454	0.072	0.9426	-1.2184	1.3113	
cp_4.0	1.9493	0.6491	3.003	0.0027	0.6770	3.2215	**
sex_1.0	1.5059	0.5135	2.933	0.0034	0.4995	2.5123	**
exang_1.0	0.7033	0.4470	1.573	0.1157	-0.1729	1.5795	
slope_2.0	0.9452	0.4589	2.060	0.0394	0.0457	1.8446	*
slope_3.0	0.7884	0.8860	0.890	0.3736	-0.9482	2.5249	
thal_6.0	-0.2374	0.7734	-0.307	0.7588	-1.7533	1.2784	
thal_7.0	1.3521	0.4212	3.210	0.0013	0.5266	2.1776	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(b) Full model summary - Custom LogReg Validation

Figure 1: Full Model Comparison R and our (Python) validation.

```
> step_model <- step(full_model, direction = "backward", trace = TRUE)
Start: AIC=224.82
target ~ age + sex + cp + trestbps + chol + fbs + restecg + thalach +
       exang + oldpeak + slope + ca + thal

  Df Deviance   AIC
- restecg    2  188.35 222.35
- age        1  187.04 223.04
- fbs        1  187.43 223.43
- chol       1  188.77 224.77
<none>      186.82 224.82
- slope      2  191.09 225.09
- exang      1  189.38 225.38
- thalach    1  189.72 225.72
- oldpeak    1  190.35 226.35
- trestbps   1  190.55 226.55
- sex        1  195.31 231.31
- thal       2  199.06 233.06
- cp         3  204.33 236.33
- ca         1  211.12 247.12
```

Step 1 - R

```
Step: AIC=220.55
target ~ sex + cp + trestbps + chol + fbs + thalach + exang +
       oldpeak + slope + ca + thal
```

```
  Df Deviance   AIC
- fbs    1  189.17 219.17
<none>   188.54 220.54
- chol   1  190.95 220.95
- exang   1  191.28 221.28
- slope   2  193.29 221.29
- thalach 1  191.47 221.47
- oldpeak 1  191.87 221.87
- trestbps 1  192.28 222.28
- thal    2  200.04 228.04
- sex     1  198.38 228.38
- cp      3  206.06 232.06
- ca      1  214.34 244.34
```

Step 3 - R

```
Step: AIC=222.35
target ~ age + sex + cp + trestbps + chol + fbs + thalach + exang +
       oldpeak + slope + ca + thal
```

```
  Df Deviance   AIC
- age    1  188.54 220.54
- fbs    1  188.97 220.97
<none>   188.35 222.35
- chol   1  190.86 222.86
- exang   1  190.95 222.95
- slope   2  193.18 223.18
- thalach 1  191.41 223.41
- oldpeak 1  191.57 223.57
- trestbps 1  192.28 224.28
- sex     1  197.72 229.72
- thal    2  199.81 229.81
- cp      3  205.67 233.67
- ca      1  213.24 245.24
```

Step 2 - R

```
Step: AIC=219.17
target ~ sex + cp + trestbps + chol + thalach + exang + oldpeak +
       slope + ca + thal
```

```
  Df Deviance   AIC
<none>   189.17 219.17
- chol   1  191.41 219.41
- slope   2  193.50 219.50
- exang   1  191.62 219.62
- thalach 1  192.30 220.30
- trestbps 1  192.59 220.59
- oldpeak 1  192.93 220.93
- sex     1  198.51 226.51
- thal    2  201.58 227.58
- cp      3  208.47 232.47
- ca      1  214.35 242.35
```

Step 4 - R

(a) R backward elimination trace

```
Start: AIC=224.8201
Variables: Intercept, cp, sex, fbs, restecg, exang, slope, thal, age, trestbps, chol, thalach, oldpeak, ca

Step: AIC=222.3511 Dropping Group: restecg

Step: AIC=220.5454 Dropping Group: age

Step: AIC=219.1728 Dropping Group: fbs

=====
Stepwise selection finished (no further AIC improvement).
Final Model AIC: 219.1728
Final Variables: Intercept, cp, sex, exang, slope, thal, trestbps, chol, thalach, oldpeak, ca
```

(b) Custom Python backward elimination trace

Figure 2: Comparing the backward elimination trace of R and our (Python) implementation.

```

> summary(step_model)

Call:
glm(formula = target ~ sex + cp + trestbps + chol + thalach +
     exang + oldpeak + slope + ca + thal, family = binomial, data = data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.675e+01  9.797e+00 -2.730  0.00633 **
sex1         1.506e+00  5.135e-01  2.933  0.00336 **
cp2          7.906e-01  7.589e-01  1.042  0.29748
cp3          4.646e-02  6.454e-01  0.072  0.94261
cp4          1.949e+00  6.491e-01  3.003  0.00267 **
trestbps     2.894e+00  1.584e+00  1.827  0.06777 .
chol         1.706e+00  1.154e+00  1.479  0.13927
thalach     -3.850e-05  2.212e-05 -1.740  0.08185 .
exang1       7.033e-01  4.471e-01  1.573  0.11568
oldpeak      4.397e-01  2.318e-01  1.897  0.05778 .
slope2       9.452e-01  4.589e-01  2.060  0.03944 *
slope3       7.884e-01  8.860e-01  0.890  0.37358
ca           1.177e+00  2.606e-01  4.516  6.31e-06 ***
thal6       -2.374e-01  7.734e-01 -0.307  0.75885
thal7       1.352e+00  4.212e-01  3.210  0.00133 **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 395.69  on 286  degrees of freedom
Residual deviance: 189.17  on 272  degrees of freedom
AIC: 219.17

Number of Fisher Scoring iterations: 6

>
> # Extract coefficients for comparison with your Python implementation
> coef(step_model)
      (Intercept)      sex1          cp2          cp3          cp4      trestbps          chol
-2.674796e+01  1.505892e+00  7.906335e-01  4.645639e-02  1.949263e+00  2.893614e+00  1.706227e+00
      thalach      exang1      oldpeak      slope2      slope3      ca      thal6
-3.849603e-05  7.032951e-01  4.397302e-01  9.451510e-01  7.883728e-01  1.176698e+00 -2.374375e-01
      thal7
1.352073e+00

```

(a) Final model summary and all coefficients - R

Variable	Coef	Std.Err	z	P> z	[0.025	0.975]	
Intercept	-26.7479	9.7965	-2.730	0.0063	-45.9486	-7.5472	**
trestbps	2.8936	1.5842	1.827	0.0678	-0.2113	5.9985	.
chol	1.7062	1.1540	1.479	0.1393	-0.5555	3.9680	
thalach	-0.0000	0.0000	-1.740	0.0818	-0.0001	0.0000	.
oldpeak	0.4397	0.2318	1.897	0.0578	-0.0145	0.8940	.
ca	1.1767	0.2606	4.516	0.0000	0.6660	1.6874	***
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cp_3.0	0.0465	0.6454	0.072	0.9426	-1.2184	1.3113	
cp_4.0	1.9493	0.6491	3.003	0.0027	0.6770	3.2215	**
sex_1.0	1.5059	0.5135	2.933	0.0034	0.4995	2.5123	**
exang_1.0	0.7033	0.4470	1.573	0.1157	-0.1729	1.5795	
slope_2.0	0.9452	0.4589	2.060	0.0394	0.0457	1.8446	*
slope_3.0	0.7884	0.8860	0.890	0.3736	-0.9482	2.5249	
thal_6.0	-0.2374	0.7734	-0.307	0.7588	-1.7533	1.2784	
thal_7.0	1.3521	0.4212	3.210	0.0013	0.5266	2.1776	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(b) Final model summary and all coefficients - Python LogReg Validation

Figure 3: Final Model and Coefficients Comparison - R and Python